xplor ACADEMY

ROBOTICS BUILD BOOK

BUILD YOUR 4WD ROBOT

A step-by-step guide to assembling the Keyestudio 4WD Bluetooth Multi-purpose Car V2.0, from the first screw to the first drive.

ROBOTICS ARDUINO HANDS-ON

Kit KS0559 · 8 build parts

WELCOME, EXPLORER

LET'S BUILD A ROBOT

Every great creator starts by putting the pieces together.

In this book you will assemble a four-wheel-drive robot car that can follow lines, avoid obstacles, show facial expressions on an LED panel, and take commands from an infrared remote or a phone over Bluetooth. No soldering is needed. Every part fits together with screws, nuts and copper pillars.

1

LOOK

Each part starts with the components you need. Find them and lay them out before you begin.

2

BUILD

Follow the installation diagrams in order. Match each screw to the size shown.

3

CHECK

Compare your build with the finished picture before you move on to the next part.

CONTENTS

01	MEET YOUR ROBOT	Features and specifications
02	WHAT'S IN THE KIT	All 38 items
03	BEFORE YOU START	Workspace, tools and safety
04	THE BUILD, PARTS 1–8	Step-by-step assembly
05	WIRING IT UP	Part 8: motors, sensors, power
06	FINAL CHECK	Before the first power-on
07	WHAT'S NEXT	17 projects to code

ONE CAR, MANY MISSIONS

A sturdy 4WD chassis with a brain, eyes and a face.

The robot is controlled by the **Keyestudio V4.0 board**, an Arduino Uno-compatible board built on the ATmega328P microcontroller. On top of it sits a **motor driver expansion board** using the DRV8833 chip, which powers the four motors and gives every sensor a plug-in port. The kit comes with 17 learning projects that grow from simple to advanced.

WHAT IT CAN DO

- Follow a black line with the 3-channel line-tracking sensor
- Measure distance and avoid obstacles with the ultrasonic sensor
- Follow your hand as you move it in front of the car
- Stay inside a boundary drawn on the floor
- Show faces and icons on the 8×16 LED board
- Drive by IR remote or by phone app (iOS and Android)

SPECIFICATIONS

WORKING VOLTAGE	5 V
INPUT VOLTAGE	6–9 V
MAX OUTPUT CURRENT	2 A
MOTOR SPEED	200 rpm at 5 V
MOTOR DRIVER	DRV8833
ULTRASONIC RANGE	2–400 cm, <15° angle
IR REMOTE RANGE	10 m (measured)
BLUETOOTH RANGE	50 m (measured)

Keyestudio V4.0 development board (image loads from docs.keyestudio.com)

The brain. 14 digital pins (6 with PWM), 6 analog inputs, 16 MHz clock, USB port and reset button.

Motor driver expansion board with its ports labelled (image loads from docs.keyestudio.com)

The muscles. Two H-bridge channels up to 1.5 A each, a built-in IR receiver, and PH2.0 ports for every sensor.

CHECK YOUR PARTS

Tick off every item before you start. Small screws are easy to lose.

#	ITEM	QTY
1	Keyestudio V4.0 board (UNO compatible)	1
2	Motor driver expansion board	1
3	Bluetooth module	1
4	Red LED module	1
5	Ultrasonic sensor	1
6	Line-tracking sensor	1
7	8×16 LED board (with 4-pin wire)	1
8	Servo	1
9	IR remote control	1
10	USB cable	1
11	18650 battery holder	1
12	6× AA battery holder	1
13	Cradle head	1
14	150 mm JST-PH2.0 5-pin wire	1
15	150 mm PH2.0 4-pin to Dupont wire	1
16	Acrylic board	1
17	Top board	1
18	Base board	1
19	Fixed parts	4

#	ITEM	QTY
20	Wheels	4
21	M3×10 mm copper pillar	10
22	M3×40 mm copper pillar	4
23	M3×30 mm round-head screw	8
24	M3×6 mm round-head screw	40
25	M3 nickel-plated nut	16
26	M2×8 mm round-head screw	6
27	M3×8 mm round-head screw	4
28	M2 nickel-plated nut	6
29	M3×10 mm flat-head screw	3
30	Motor	4
31	Screwdriver	1
32	Black cable tie	6
33	Winding pipe	1
34	20 cm 3-pin F-F Dupont cable	1
35	Decorative card	1
36	Blue jumper cap	4
37	Red jumper cap	4
38	Tracking runway	1

SET UP FOR SUCCESS

Ten minutes of preparation saves an hour of searching for screws.

YOUR WORKSPACE

- A clear, flat table with good light
- A small tray or bowl for each screw size
- The screwdriver from the kit
- About 60–90 minutes for the full build
- A computer with a USB port, for Part 4

KNOW YOUR SCREWS

- **M3 vs M2:** M2 is the thinner screw. It is only used in a few places.
- **The second number is the length.** An M3×6 is short, an M3×30 is long.
- **Pillars** are the gold hex spacers that hold boards apart.
- Lay one of each next to the kit list to compare.

HOW EVERY PART WORKS

COMPONENTS
Collect exactly the pieces in the picture.

INSTALLATION
Follow the diagram. Arrows show where each piece goes.

PROTOTYPE
Check that your build matches before moving on.

TIP Tighten screws until they are snug, then stop. Over-tightening can crack the acrylic board or strip the plastic.

SAFETY Keep batteries out of the holder until the build is finished. Check the + and − markings every time you insert them. Ask an adult to help with 18650 lithium batteries.

TEAMWORK Building in pairs works well: one person reads the step and finds the parts, the other assembles. Swap roles after each part.

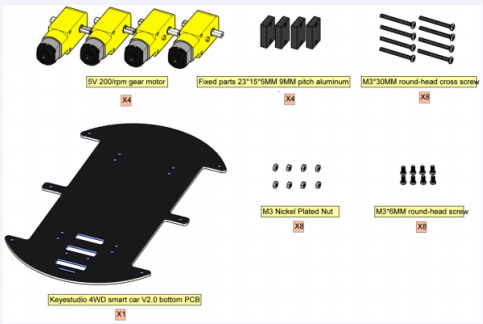
PART 1

THE BUILD

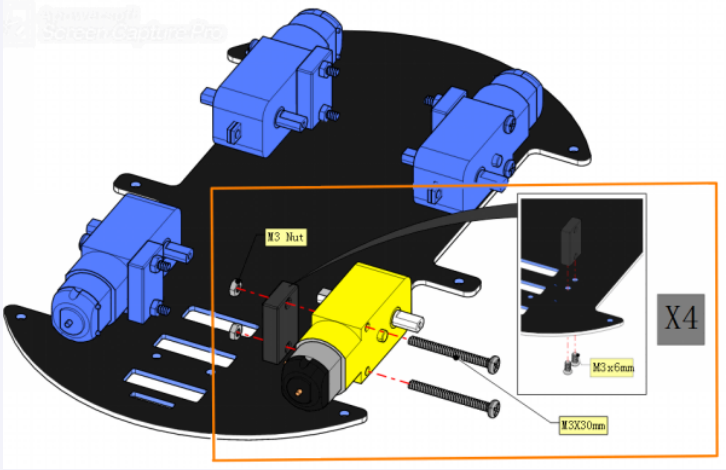
PART 1

The first pieces of your chassis.

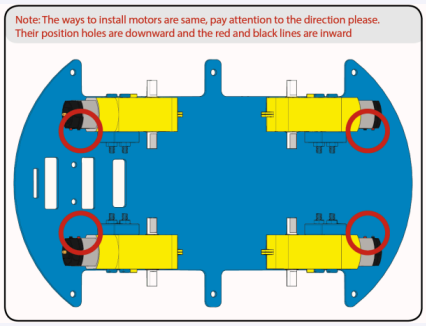
1.1 COMPONENTS NEEDED



1.2 INSTALLATION DIAGRAM



1.3 CHECK YOUR PROTOTYPE



TIP Count the screws in the components picture and set exactly that many aside. If you have some left over at the end of the step, look again at the diagram.

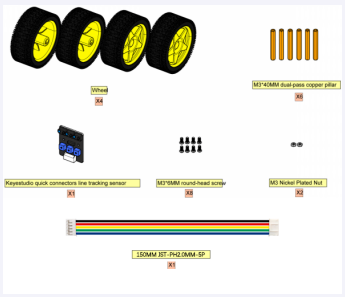
PART 2

THE BUILD

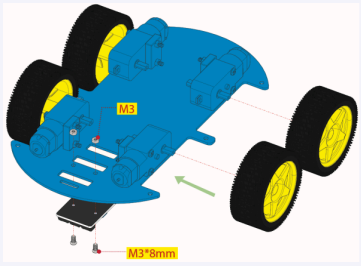
PART 2

Building on the base.

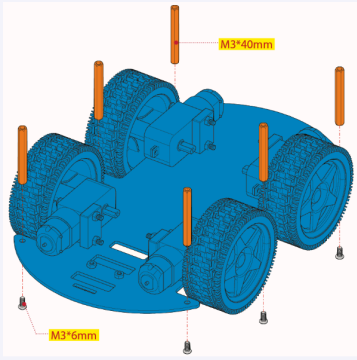
2.1 COMPONENTS NEEDED



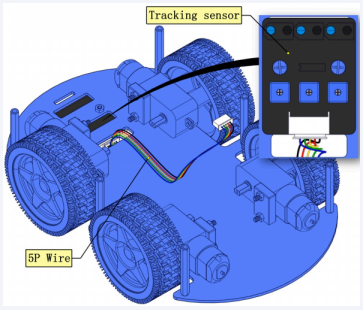
2.2 INSTALLATION DIAGRAM



Step A

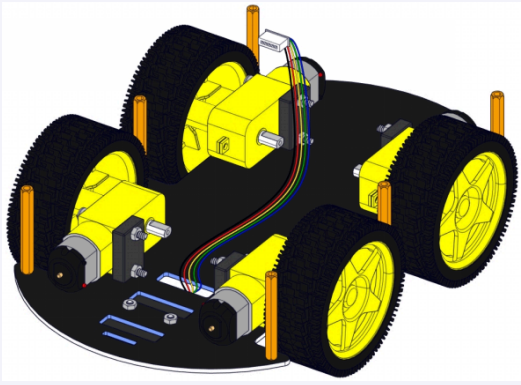


Step B



Step C

2.3 CHECK YOUR PROTOTYPE



CHECK Turn the chassis over and look at it from every side. Each piece should sit flat with no gaps.

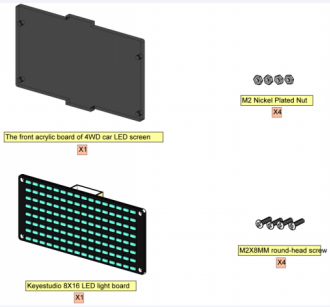
PART 3

THE BUILD

PART 3

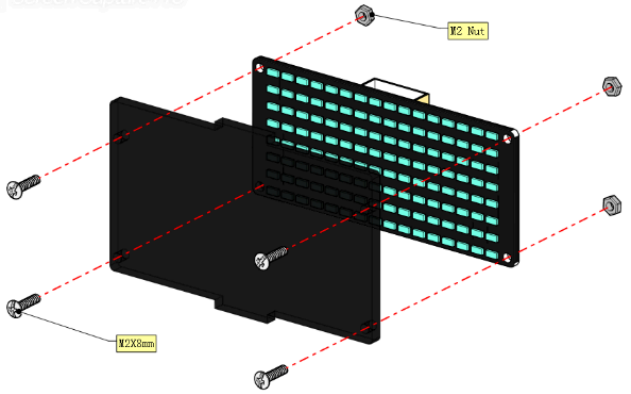
Mounting and plugging in the first wire.

3.1 COMPONENTS NEEDED

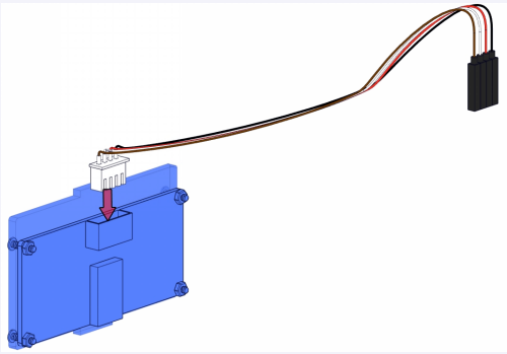




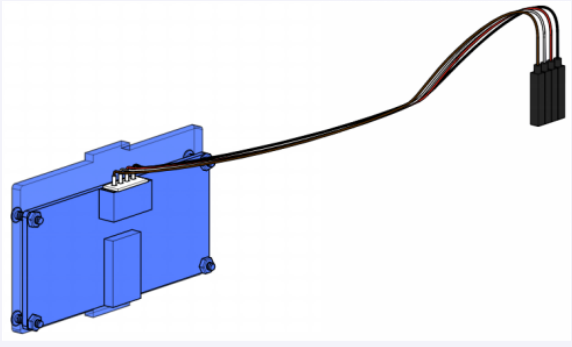
3.2 INSTALLATION DIAGRAM



3.3 PLUG IN THE WIRE



Plug wire



3.4 · Check your prototype

TIP PH2.0 plugs only fit one way. If a plug won't go in, flip it over instead of pushing harder.

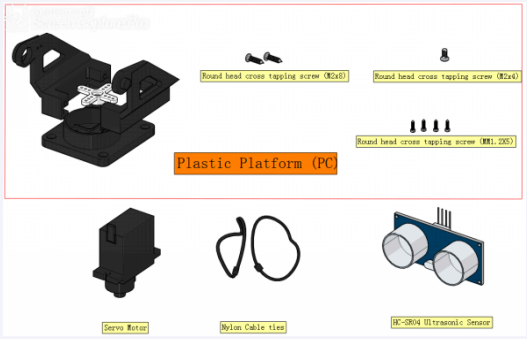
PART 4

THE BUILD

PART 4 · SERVO

Set the servo to 90° before you fit it.

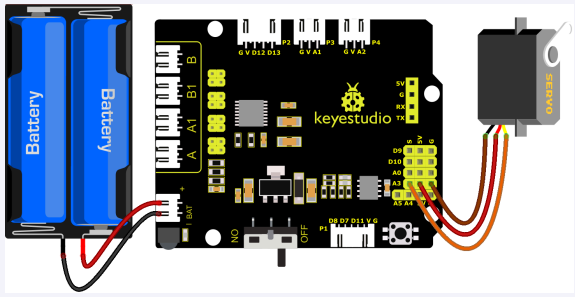
4.1 COMPONENTS NEEDED



4.2 ROTATE THE SERVO TO 90°

The servo turns the ultrasonic sensor left and right. Setting it to its middle position (90°) first means the sensor will face straight ahead once it is mounted. Connect the servo to the board, plug the board into your computer, and upload the servo code.

Servo	8833 Motor Driver Expansion Board
Brown	G
Red	5V
Yellow	A3



OPTION A · ARDUINO

Download the code: Arduino.7z. Open the servo sketch in the Arduino IDE and upload it.

OPTION B · KIDSBLOCK

Download the code: Kidsblock.7z. Open the servo project in KidsBlock and upload it.

IMPORTANT Choose one option and follow the matching course. If your computer doesn't recognise the board, install the USB driver first (Keyestudio guide, chapter 3).

FIT THE SERVO

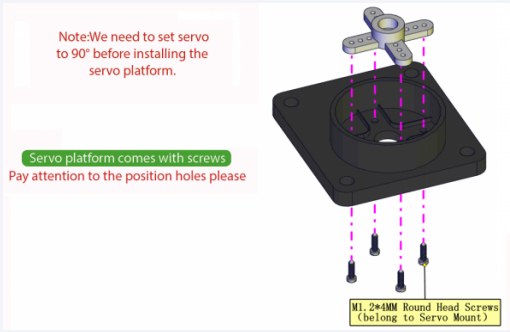


Arduino servo code

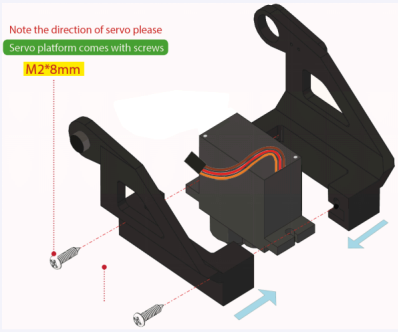


KidsBlock servo code

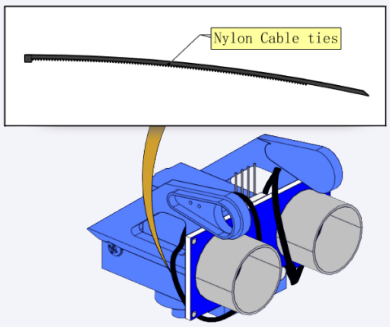
4.3 INSTALLATION DIAGRAM



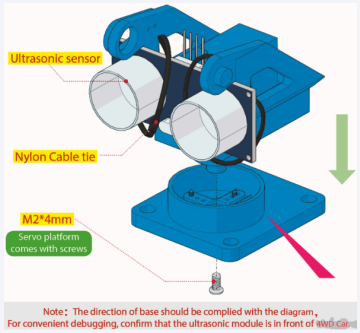
Step A



Step B

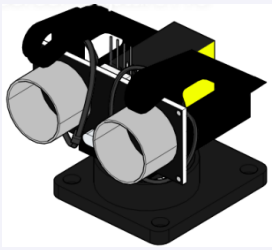


Step C



Step D

4.4 CHECK YOUR PROTOTYPE



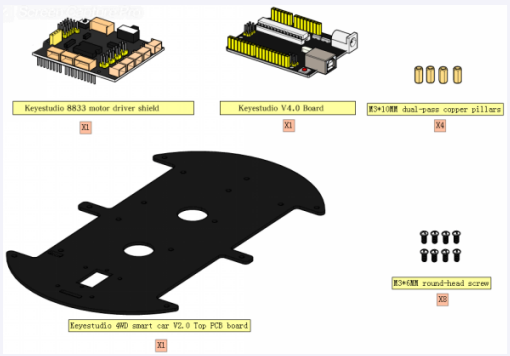
PART 5

THE BUILD

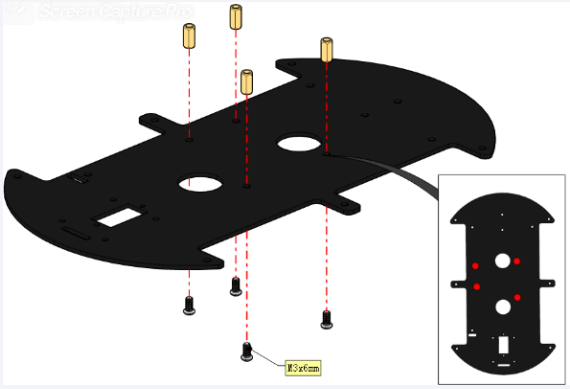
PART 5

Past halfway. Keep going.

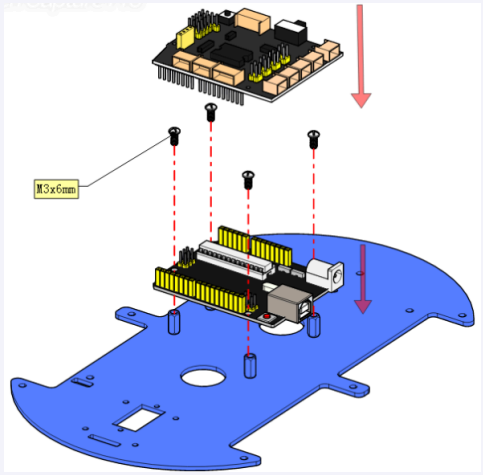
5.1 COMPONENTS NEEDED



5.2 INSTALLATION DIAGRAM

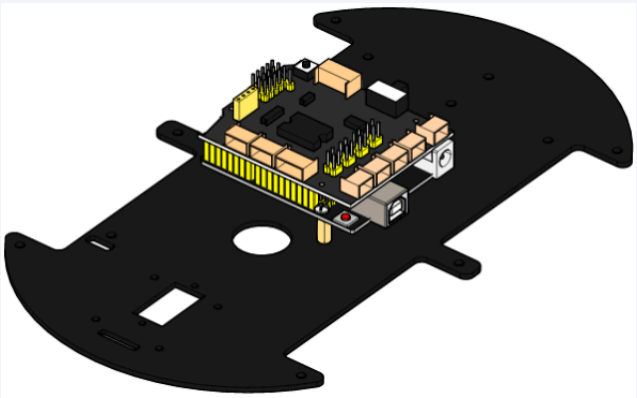


Step A



Step B

5.3 CHECK YOUR PROTOTYPE



PART
6

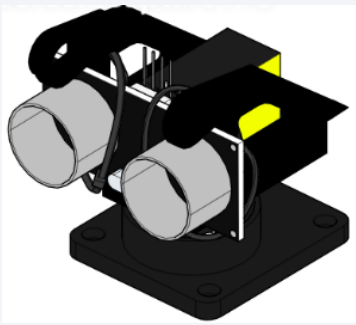
THE BUILD

PART 6

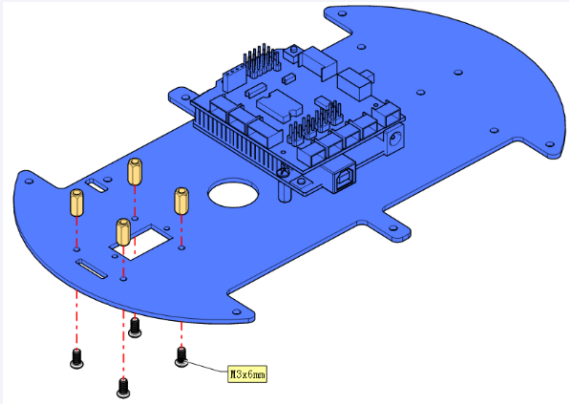
Adding the next layer.

6.1 COMPONENTS NEEDED

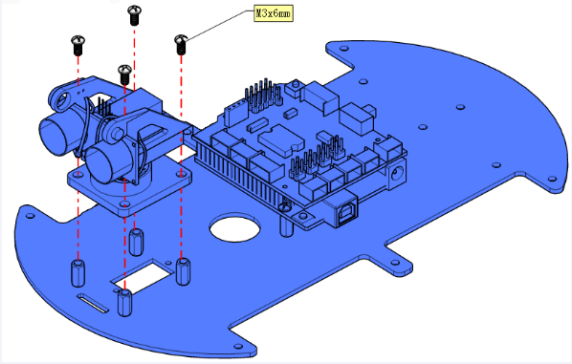




6.2 INSTALLATION DIAGRAM

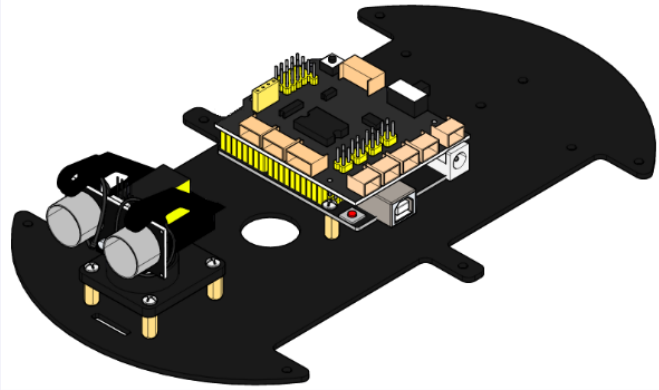


Step A



Step B

6.3 CHECK YOUR PROTOTYPE



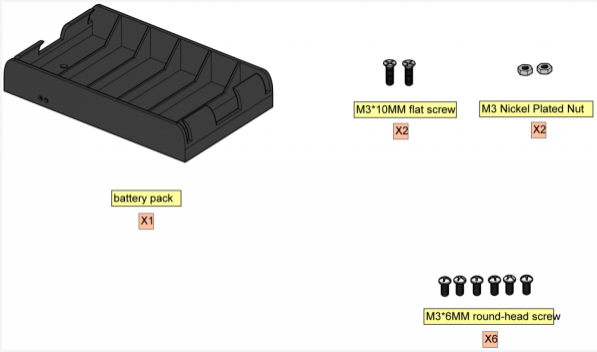
PART 7

THE BUILD

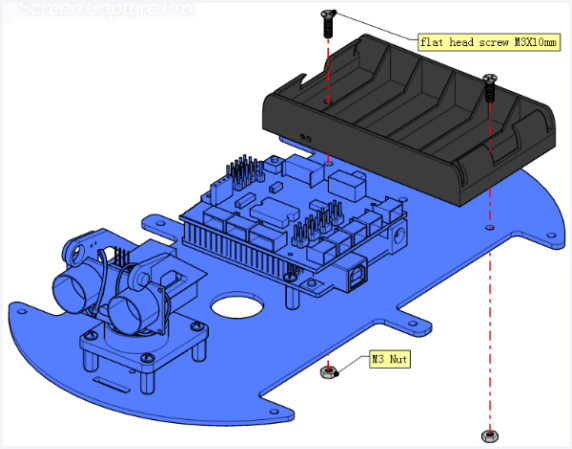
PART 7

The last mechanical step before wiring.

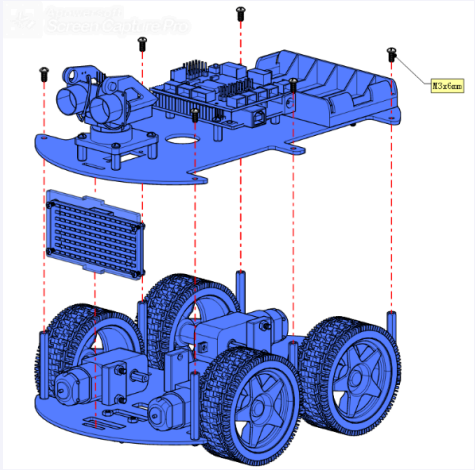
7.1 COMPONENTS NEEDED



7.2 INSTALLATION DIAGRAM

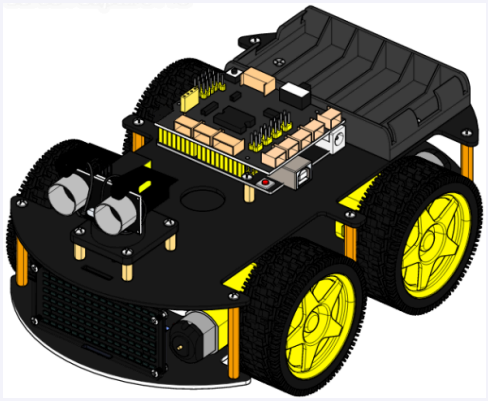


Step A



Step B

7.3 CHECK YOUR PROTOTYPE

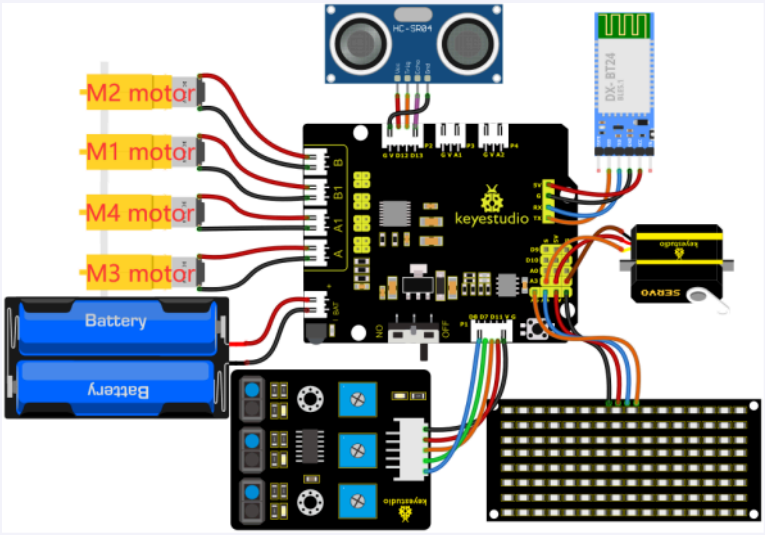


PART 8

PART 8 · WIRING

Connect the motors, sensors and power.

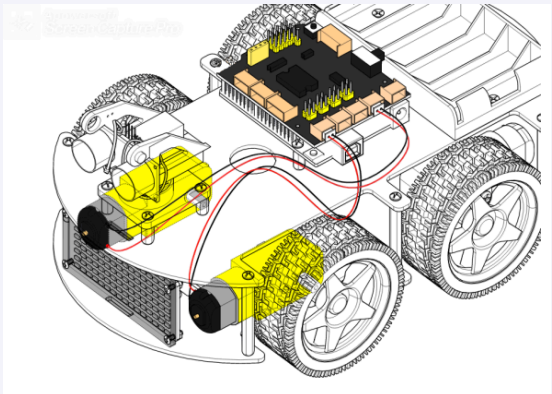
8.1 WIRING DIAGRAM



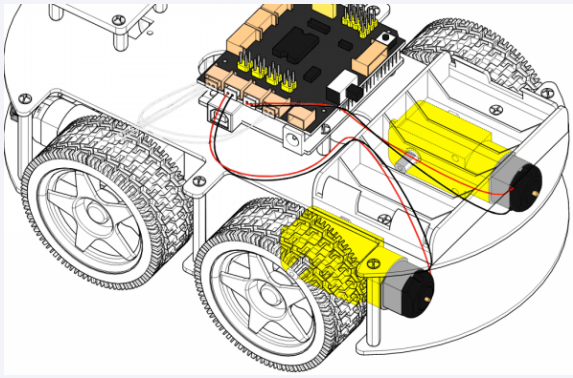
MOTOR CONNECTIONS

Motor M1	→ B1
Motor M2	→ B
Motor M3	→ A
Motor M4	→ A1

8.2 CONNECT THE MOTORS

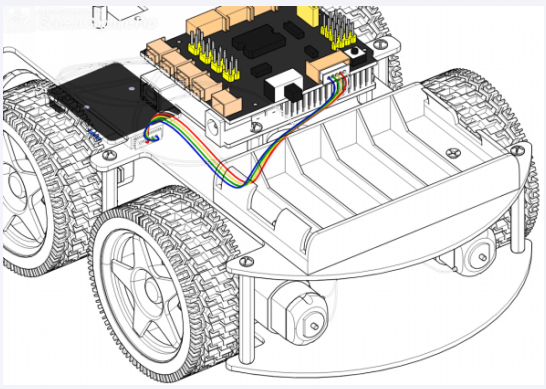


8.2 · M2 → B, M3 → A

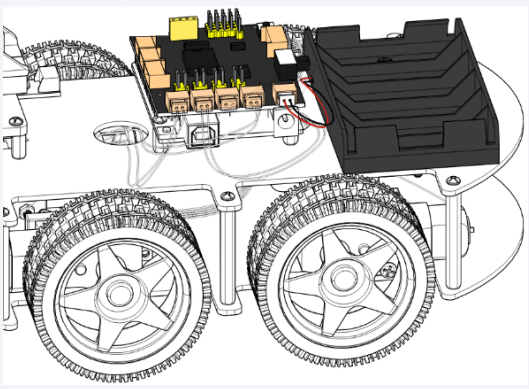


8.3 · M1 → B1, M4 → A1

SENSORS AND POWER



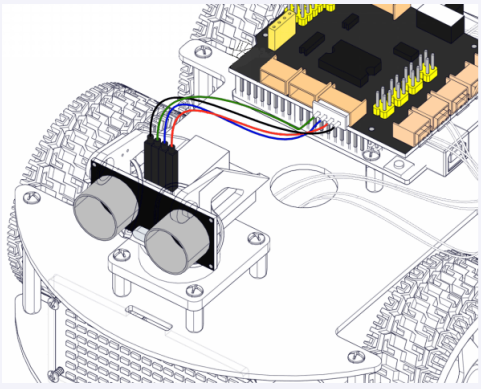
8.4 · 3-channel line-tracking sensor



8.5 · Power supply → BAT

Ultrasonic Sensor		8833 Motor Driver Expansion Board
Vcc		V
Trig		D12
Echo		D13
Gnd		G

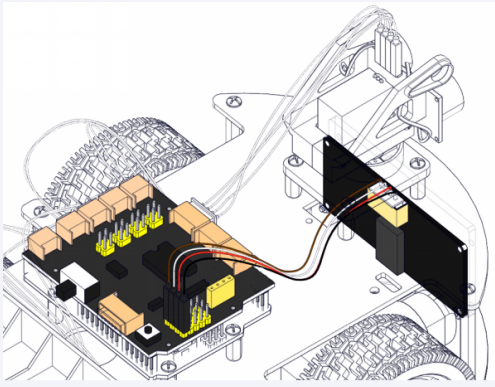
8.6 · Ultrasonic sensor



8.6 · Ultrasonic sensor (continued)

8*16 LED Board		8833 Motor Driver Expansion Board
GND		G
VCC		5V
SDA		A4
SCL		A5

8.7 · 8×16 LED board

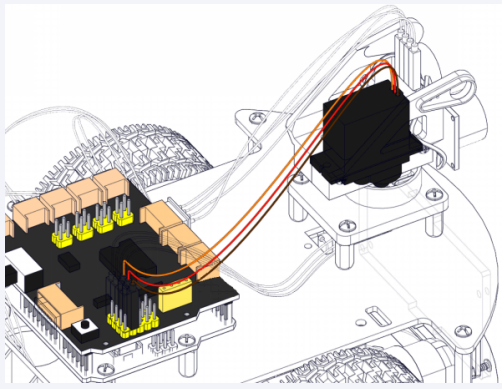


8.7 · 8×16 LED board (continued)

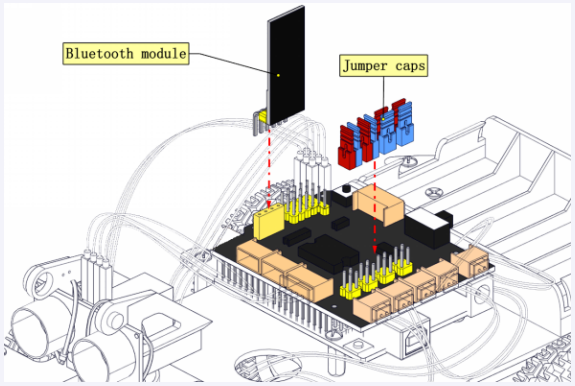
SERVO, BLUETOOTH AND FINISH

Servo	8833 Motor Driver Expansion Board
Brown	G
Red	5V
Yellow	A3

8.8 · Servo

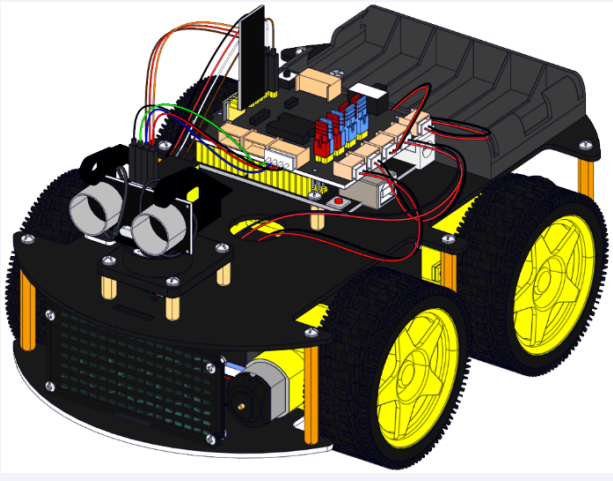


8.8 · Servo (continued)



8.9 · Plug in the Bluetooth module and jumper caps

IMPORTANT **Remove the Bluetooth module before you upload any code**, or the upload will fail. Plug it back in once the upload has finished.



8.10 · Complete prototype

READY FOR LAUNCH?

Go through this list before you put batteries in.

BUILD

- [ ] All four wheels turn freely and don't rub the chassis
- [ ] Every board sits flat on its pillars
- [ ] No screws are loose and none are left over
- [ ] The ultrasonic sensor faces straight ahead

WIRING

- [ ] M1 → B1, M2 → B, M3 → A, M4 → A1
- [ ] Line-tracking, ultrasonic, LED board and servo are plugged in
- [ ] Battery lead is in the BAT port
- [ ] Wires are tied back clear of the wheels

FIRST POWER-ON

1. Insert the batteries, checking + and − against the holder.
2. Remove the Bluetooth module, connect the USB cable and upload your first project.
3. Unplug USB, refit the Bluetooth module and switch on.
4. Lift the car off the table the first time the motors run, so it can't drive off the edge.

IF SOMETHING ISN'T WORKING

WHAT YOU SEE	WHAT TO CHECK
Code won't upload	Take out the Bluetooth module. Check the driver is installed and the right port is selected.
A wheel spins the wrong way or not at all	Check that motor's plug is in the port listed in Part 8.
Sensor looks sideways	Re-run the 90° servo code from Part 4 and refit the cradle head.
Nothing turns on	Check battery direction, battery charge and the BAT connection.

NOW BRING IT TO LIFE

Your robot is built. Next, you teach it what to do.

There are 17 projects to code, starting with a single blinking LED and ending with a multi-purpose Bluetooth car. You can work through them in two ways.

ARDUINO IDE

Write real C code. Good for learners who are ready to type their programs line by line.

KIDSBLOCK

Drag and drop code blocks. A great first step into programming, with the same 17 projects.

01	LED blink
02	Adjust LED brightness
03	Line-tracking sensor
04	Servo control
05	Ultrasonic sensor
06	IR reception
07	Bluetooth remote control
08	Motor driving and speed control
09	Facial expression LED board
10	Restricting smart car
11	Line-tracking smart car
12	Ultrasonic following smart car
13	Obstacle-avoidance smart car
14	IR remote control smart car
15	Bluetooth control smart car
16	Bluetooth speed control smart car
17	Multi-purpose Bluetooth smart car

RESOURCES Code files, drivers and project guides are in the Keyestudio KS0559 documentation: docs.keyestudio.com/projects/KS0559

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